

- ✓ The assignment must be submitted within the deadline. **Late submissions will be penalized.**
- ✓ Submission of softcopy of the assignment is mandatory.
- ✓ Submit your assignment through this link - <https://forms.gle/7e92z5kYnS5D8y9e9>

■ Question 1 of 4 [12 marks]

Find the fundamental time periods (N/T) and the corresponding fundamental frequencies (Ω_o, ω_o) for the given signals. Also figure out the order of harmonics (if applicable) for additional components in the signal.

- (a) [3 marks] $5 + \sin(16\pi t + 30^\circ) + 6 \cos(40\pi t - 45^\circ) + 9 \sin(4\pi t + 15^\circ) - \cos(4\pi t + 50^\circ)$
- (b) [3 marks] $7.3 \sin(\frac{7}{13}t + 37^\circ) + 1.9 \cos(\frac{63}{13}t - 13^\circ)$
- (c) [3 marks] $5 + 3 \sin(\frac{3\pi}{14}n + 30^\circ) - 5 \cos(\frac{5\pi}{21}n - 15^\circ) + 7 \sin(\frac{7\pi}{42}n - 35^\circ)$
- (d) [3 marks] $3 + \sin(\frac{5\pi}{101}n + 31^\circ) + 11 \cos(\frac{31}{202}n - 13^\circ) + 12 \cos(\frac{11\pi}{303}n + 71^\circ)$

■ Question 2 of 4 [10 marks]

Find the magnitude and the phase plots for the spectrum of the given signals.

- (a) [5 marks] $5 + \sin(16\pi t + 30^\circ) + 6 \cos(40\pi t - 45^\circ) + 9 \sin(4\pi t + 15^\circ) - \cos(8\pi t + 50^\circ)$
- (b) [5 marks] $5 + 3 \sin(\frac{3\pi}{14}n + 30^\circ) - 5 \cos(\frac{5\pi}{21}n - 15^\circ) + 7 \sin(\frac{7\pi}{42}n - 35^\circ) + 14 \cos(\frac{15\pi}{21}n - 45^\circ)$

Note - You have to maintain appropriate ranges.

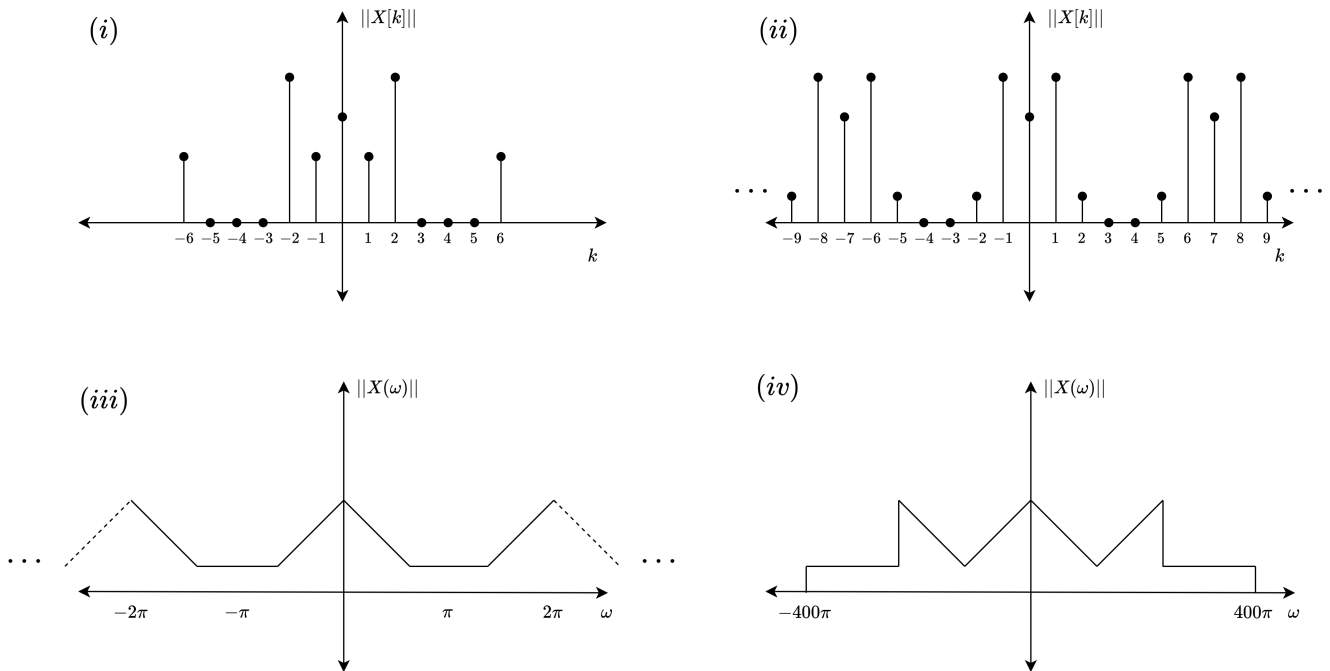
■ Question 3 of 4 [8 marks]

We have talked about how the kernel length indicated how good a filter response is before the mid. We now have the means to see it in action. Considering the standard LPF kernel is of the form $\frac{1}{3} \cdot [1, \underline{1}, 1]$ where the length is 3, apply DTFT on the LPF kernel for length 3, 5, 7, and 9. Plot the magnitude graphs and comment on your findings. Use appropriate frequency ranges. For the plotting you can take help of Desmos.

Note - Underlined term represents the 0 point of the signal.

Also note - You will not be asked to plot figures of lengthy functions in the final. You will be given small manageable kernels.

■ Question 4 of 4 [15 marks]



- (a) [8 marks] Based on the characteristics of the given spectrums, comment on the probable characteristics of their time domain counterparts.
- (b) [7 marks] If the time domain representation of the CT aperiodic signal according to your answer in part (a) is sampled at a sampling frequency of 800 Hz, what would be the corresponding spectrum? Draw the spectrum.